

# **Solving Systems of Linear Equations**

**ECON 441: Introduction to Mathematical Economics**  
**Lecture 7**

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# A Simple Economic Model

Two equations in two unknowns:

$$q + 2p = 100$$

$$q - 3p = 20$$

Can write this as:

$$Ax = b$$

where

$$A = \begin{bmatrix} 1 & 2 \\ 1 & -3 \end{bmatrix} \quad x = \begin{bmatrix} q \\ p \end{bmatrix} \quad b = \begin{bmatrix} 100 \\ 20 \end{bmatrix}$$

## Solution using Matrix Inversion

$$A = \begin{bmatrix} 1 & 2 \\ 1 & -3 \end{bmatrix} \quad x = \begin{bmatrix} q \\ p \end{bmatrix} \quad b = \begin{bmatrix} 100 \\ 20 \end{bmatrix}$$

# Cramer's Rule

More efficient way of solving a system of equations

The  $k$ th element of  $x$  can be solved by:

$$x_k^* = \frac{|A_k|}{|A|}$$

where  $A_k$  is a matrix formed by exchanging  $k$ th column of  $A$  by  $b$ .

## Solution using Cramer's Rule

$$A = \begin{bmatrix} 1 & 2 \\ 1 & -3 \end{bmatrix} \quad x = \begin{bmatrix} q \\ p \end{bmatrix} \quad b = \begin{bmatrix} 100 \\ 20 \end{bmatrix}$$

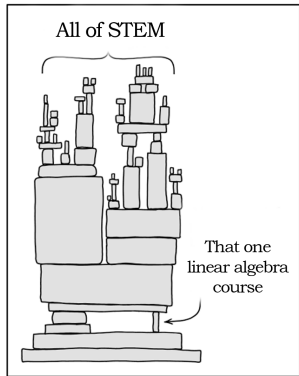
# Homogeneous equation system

A homogeneous equation system is given by

$$Ax = 0$$

If  $A$  is nonsingular,  $x^* = A^{-1}0 = 0$ .

If  $A$  is singular there can be infinite number of solutions (this is true for any system of equations).



# Network Theory

A network of connections can be expressed as an adjacency matrix.

$$M = \begin{pmatrix} m_{11} & m_{12} & \cdots & m_{1n} \\ m_{21} & m_{22} & \cdots & m_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ m_{n1} & m_{n2} & \cdots & m_{nn} \end{pmatrix}$$

where

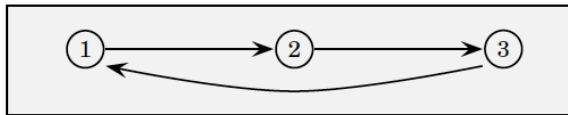
$$m_{ij} = \begin{cases} 1 & \text{if there is a direct link from } i \text{ to } j \\ 0 & \text{otherwise} \end{cases}$$

# Network Theory

Consider the following network:

$$M = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad M^2 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad M^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The matrices  $M, M^2, M^3$  give the nodes reachable in one, two and threesteps from any initial node.



# Network Theory

Consider the sum:

$$S_k = M + M^2 + M^3 + \dots + M^k$$

The  $(i, j)$  element of  $S_k$  gives the number of paths of length  $k$  or less, from  $i$  to  $j$ .

For the previous example:

$$S_3 = M + M^2 + M^3 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

So there is one way to go from any node to any other in three or fewer steps.

# Uses of Network Theory

- Network theory can be used to model
  - Interconnectedness of financial institutions (to predict risk of banking collapses)
  - Interconnectedness of the countries in world trade
  - Predicting supply chain risk

# Markov Chain

A Markov Chain can model the transition between different states.

Example: Employment (E) and Unemployment (U).

Transition matrix:

$$P = \begin{pmatrix} P(E \rightarrow E) & P(U \rightarrow E) \\ P(E \rightarrow U) & P(U \rightarrow U) \end{pmatrix} = \begin{pmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{pmatrix}$$

Say the initial state vector is:

$$\pi(0) = \begin{bmatrix} \pi_E(0) \\ \pi_U(0) \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}$$

# Transition Matrix

After one period, the state distribution is:

$$\pi(1) = \begin{bmatrix} \pi_E(1) \\ \pi_U(1) \end{bmatrix} = P\pi(0) = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 0.76 \\ 0.24 \end{bmatrix}$$

After  $t$  periods:

$$\pi(t) = P^t \pi(0)$$

# Ordinary Least Squares

Linear model with  $k$  variables:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_k X_{ik} + \varepsilon_i$$

where  $i = 1, \dots, n$  denotes  $n$  observations.

Denote

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}_{n \times 1}, \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}_{k \times 1}, X = \begin{bmatrix} 1 & X_{11} & \cdots & X_{1k} \\ 1 & X_{12} & \cdots & X_{2k} \\ \vdots & \vdots & \cdots & \vdots \\ 1 & X_{1n} & \cdots & X_{nk} \end{bmatrix}_{n \times k}, \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}_{n \times 1}$$

Then,

$$Y = X\beta + \varepsilon \quad \text{OLS estimator: } \hat{\beta} = (X^T X)^{-1} X^T Y$$

# Natural Language Processing (NLP)

Bag of Words (BoW) model is a simple and widely used method in NLP  
Transform text into fixed-length vectors by counting how many times each word appears in a document

Example:

- Doc1: “the cat sat on the mat”
- Doc2: “the dog sat on the log”
- Vocabulary for these documents: [*the, cat, sat, on, mat, dog, log*]
- Vector for Doc1: [2, 1, 1, 1, 1, 0, 0]
- Vector for Doc2: [2, 0, 1, 1, 0, 1, 1]

## What's next?

Midterm 1 is next week: review on Monday, exam on Wednesday

Sample exams and a help sheet are posted on the Exams page

Notes for reviewing Linear Algebra are on the website (not a substitute for the textbook for understanding concepts)

After the midterm, we move on to differential calculus

# Homework Questions

- Exercise 5.4: 6
- Exercise 5.5: 1, 2, 3 (a), (d)

Textbook reference: 5.4, 5.5, 4.7